

Introduction to R

Session 1

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2026-10-08

Welcome to “Intro to R”

- Two sessions:
 - Thursday, October 8th, 1:00pm - 4:00pm
 - Friday, October 9th, 9:30am - 12:30pm
- Course website:
 - <https://github.com/aaslinur/intro-to-R>
 - Slides, exercises, and solutions

Instructors

- **Instructor**

- **Aasli Abdi Nur**

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- **Teaching Assistants**

- **Laura Leone**, DPhil Student
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Course Goals

- Overview
 - Describe why **R** is a powerful tool for social science research
- Installation
 - Successfully install **R** and **R Studio**

Course Agenda

- **Session 1**

- Module 1: Introduction to [R](#), [RStudio](#), and code formats
- Module 2: [R](#) programming fundamentals (syntax, operators, data types, data structures, sequencing)
- Module 3: Working with data (indexing vectors / matrices, importing data)

- **Session 2**

- Module 4: Importing and exporting data
- Module 5: Data manipulation ([dplyr](#)) and data visualization ([ggplot2](#))
- Module 6: Best practices and resources for self-study

Module 1

Introduction to R, RStudio, and Code Formats

Learning objectives:

- Installing R and RStudio
- Exploring the benefits of R and R Studio
- Understanding R Scripts, R notebooks, Quarto documents

Installing **R** and **RStudio**

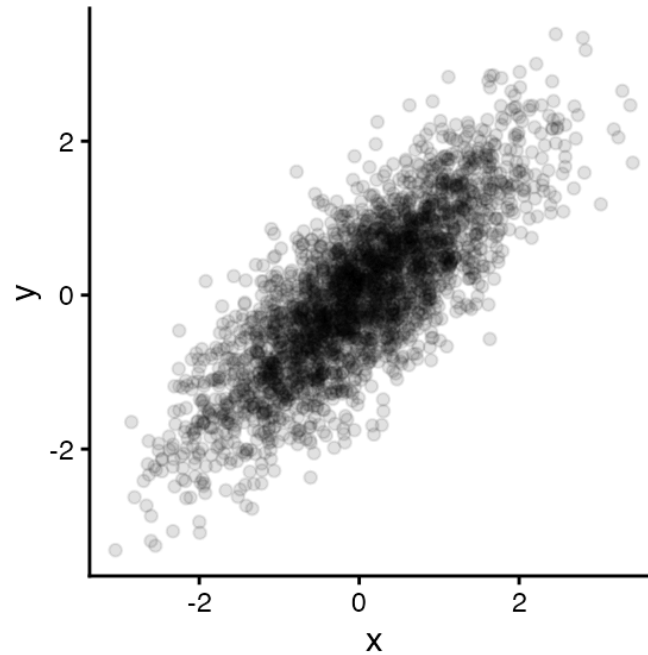
- **R** is a statistical programming language
 - Download: <https://cloud.r-project.org>
- **RStudio** is an integrated development environment (IDE) for **R** programming
 - Download: <https://posit.co/downloads/>

Exploring the benefits of **R**?

- **Cost:** free and open source - you don't need a terminal server or license.
- **Flexibility:** can handle virtually any data format.
- **Functionality:** powerful language for data manipulation, statistical analysis, and publication-ready data visualizations
- **Reproducibility:** makes replication easy.
- **Resources:** large community for support and learning.

Easy to simulate and plot data

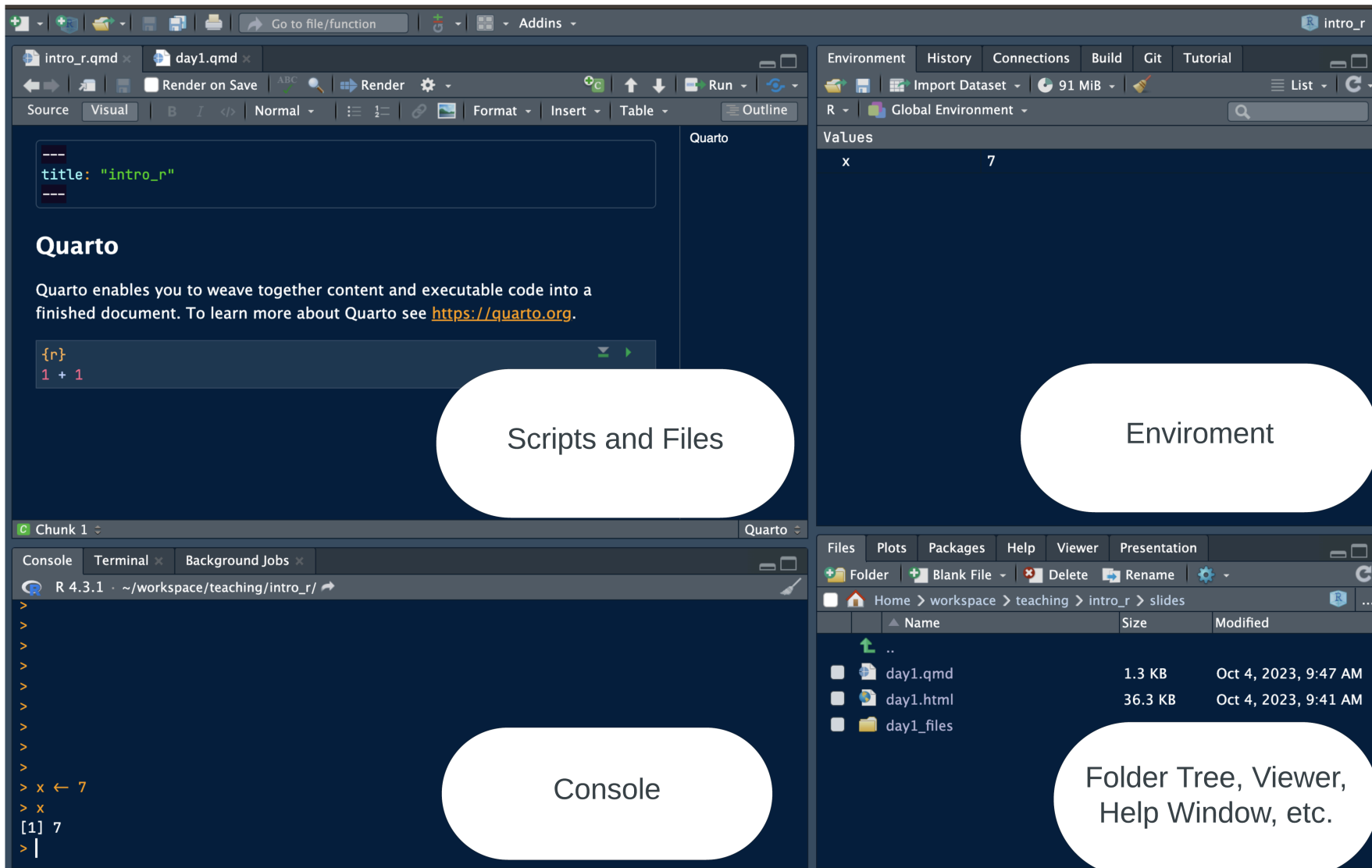
```
1 # Generate random data for x
2 x <- rnorm(n = 3000)
3 y <- 0.8 * x + rnorm(3000, 0, sqrt(1 - 0.8^2))
4
5 # Create data.frame
6 data_df <- data.frame(x = x, y = y)
7
8 # Generate visualization
9 data_df %>%
10   ggplot(aes(x = x, y = y)) +
11   geom_point(alpha = 0.1) +
12   theme_classic()
```



Why **RStudio**?

- **All-in-one development environment:** streamlines coding, data visualization, and workflow
- **Extensible:** supports R — but also Python, SQL, and Git
- **Rich community:** eases learning and problem-solving

RStudio panes

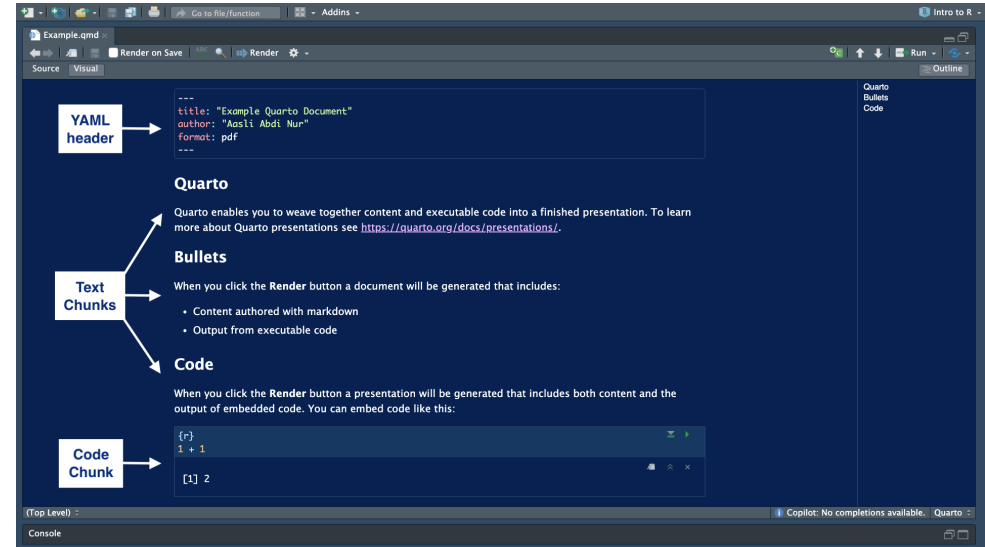


Code formats: **R** Scripts vs. **R** Notebooks

- **R** Scripts
 - Simple: just code
 - Best for simple tasks (and multi-script pipelines)
- **R** Notebooks (Quarto, **R** Notebook, **R** Markdown)
 - Integrated: Mix of code, text, and outputs for easy documentation
 - Interactive: real-time code execution and output display

Quarto documents

- “Notebook” Style: supports interactive code and text
 - Code cells: segments for code execution
 - Text chunks: annotations or explanations in Markdown format.
- Inline output: figures and code output display directly below the corresponding code cell



Installing packages

- Packages: pre-built code and functions.
- Packages are generally installed from the Comprehensive R Archive Network (CRAN)

Install: new packages

```
1 install.packages("tidyverse")
```

Library: load installed packages

```
1 library(tidyverse)
```

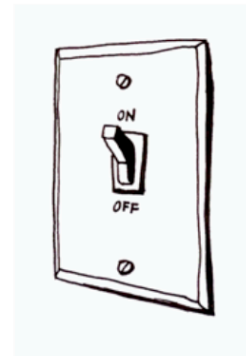
Installing a package

```
install.packages('my.package')
```



Loading a package

```
library('mypackage')
```



YaRrr! The Pirates Guide to R. Nathaniel D. Phillips, 2018.

Running code

- Run all code in a quarto document (or **R** script, or **R** notebook)
 - Exception: install packages, quick checks in console
- To run a single line of code in a code cell
 - Cursor over line, **Ctrl + Enter** (Windows/Linux) or **Cmd + Enter** (Mac).
- To run a full code cell (or script)
 - **Ctrl + Shift + Enter** (Windows/Linux) or **Cmd + Shift + Enter** (Mac).

Live coding demo

- Demo of creating a new Quarto document and running code in a code cell
- Your turn next...

In-class exercise 0

- Create a new quarto document
 - File -> New File -> Quarto Document -> Create
- Create a new code cell
 - Insert -> Executable cell -> R
- Practice running code below:

```
1 3+3
```

```
[1] 6
```

```
1 print("Thank you for attending the first session!")
```

```
[1] "Thank you for attending the first session!"
```

Module 2

R programming fundamentals

Learning objectives:

- Comprehend R objects and functions
- Master basic syntax, including comments, assignment, and operators
- Understand data structures and types in R

Objects

- Everything in R is an object
 - **Vectors:** Ordered collection of same type
 - **Data Frames:** Table of columns and rows
 - **Function:** Reusable code block
 - **List:** Ordered collection of objects

```
1  ## Objects in R
2
3  ## Numeric like `1`, `2.5`
4  x <- 2.5
5
6  ## Character: Text strings like `"hello"`
7  y <- "hello"
8
9  ## Boolean: `TRUE`, `FALSE`
10 z <- TRUE
11
12 ## Vectors
13 vec1 <- c(1, 2, 3)
14 vec2 <- c("a", "b", "c")
15
16 ## data.frames
17 df <- data.frame(vec1, vec2)
```

Functions

- Built-in “base” functions

```
1 ## Functions in R
2 result_sqrt <- sqrt(25)
3 result_sqrt
```

```
[1] 5
```

- Custom, user-defined functions

```
1 # User-Defined Functions: Custom functions
2 my_function <- function(a, b) {
3   return(a^2 + b)
4 }
5
6 my_function(2, 3)
```

```
[1] 7
```

- Functions from packages

```
1 # User-Defined Functions: Custom functions
2
3 library(here) ## library package here
4 here() ## run custom "here" function to print out working directory
```

```
[1] "/Users/sfos0581/Documents/GitHub/intro-to-R"
```

Comments

- Use `#` to start a single-line comment
- Comments are an important way to document code

```
1  ## Add comments
2
3  x <- 7 # assigns 1 to x
4
5  ## the line below won't assign 12 to x because it's commented out
6  # x <- 12
7
8  x
```

```
[1] 7
```

Assignment operators

- Use `<-` or `=` for assignment
 - `<-` is preferred and advised for readability
- Formally, assignment means “assign the result of the operation on the right to object on the left”

```
1  ## Add comments
2
3  x <- 7 # assigns 7 to x
4
5  ## Question: what does this do?
6  y <- x
```

Arithmetic operators

- Addition / Subtraction

```
1 ## R as a calculator (# adds a comment)
2 ## Addition
3 10 + 3
```

```
[1] 13
```

```
1 ## Subtraction
2 4 - 2
```

```
[1] 2
```

- Multiplication / division

```
1 ## Multiplication
2 4 * 3
```

```
[1] 12
```

```
1 ## Division
2 12 / 6
```

```
[1] 2
```

- Exponents

```
1 ## exponents
2 10^2 ## or 10 ** 2
```

```
[1] 100
```

Comparison and logical operators

Operators

Operator	Symbol
AND	&
OR	
NOT	!
Equal	==
Not Equal	!=
Greater/Less Than	> or <
Greater/Less Than or Equal	>= or <=
Element-wise In	%in%

Examples

```
1 ## Logical operators
2
3 10 == 10
```

```
[1] TRUE
```

```
1 9 == 10
```

```
[1] FALSE
```

```
1 9 < 10
```

```
[1] TRUE
```

```
1 "apple" %in% c("bananas", "oranges")
```

```
[1] FALSE
```

```
1 "apple" %in% "bananas" | "apple" %in% "apple"
```

```
[1] TRUE
```

```
1 "apple" %in% "bananas" & "apple" %in% "apple"
```

```
[1] FALSE
```

Data structures

- There are lots of data structures; we'll focus on **vectors** and **data frames**.
 - **Vectors**: One-dimensional arrays that hold elements of a single data type (e.g., all numeric or all character).
 - **Data frames**: Two-dimensional tables where each column can have a different data type; essentially a list of vectors of equal length.

Vectors and data frames

- Vector example

```
1 ## Vector Example
2 vec_example <- c(1, 2, 3, 4, 5)
3
4 vec_example ## prints out vec_example
```

```
[1] 1 2 3 4 5
```

- Data frame example

```
1 # Data.frame example
2 example_df <- data.frame(
3   ID = c(1, 2, 3, 4),
4   Name = c("Alice", "Bob", "Charlie", "David"),
5   Age = c(25, 30, 35, 40),
6   Score = c(90, 85, 88, 76)
7 )
8
9 example_df ## prints out df_example
```

	ID	Name	Age	Score
1	1	Alice	25	90
2	2	Bob	30	85
3	3	Charlie	35	88
4	4	David	40	76

Data types

- Each **vector** or **data frame** column can only contain one data type:
 - **Numeric**: Used for numerical values like integers or decimals.
 - **Character**: Holds text and alphanumeric characters.
 - **Logical**: Represents binary values - TRUE or FALSE.
 - **Factor**: Categorical data, either ordered or unordered, stored as levels.

```
1 ## generate vectors
2 vec1 <- c(1, 2, 3)
3 vec2 <- c("a", "b", "c")
4
5 ## check type
6 class(vec1)
```

```
[1] "numeric"
```

```
1 class(vec2)
```

```
[1] "character"
```

NA (missing) values in R

- **NA** represents missing or undefined data.
 - Can vary by data type (e.g., **NA_character_** and **NA_integer_**)
- **NA** values can affect summary statistics and data visualization.
- What happens when you run the code below?

```
1 vec <- c(1, 2, 3, NA)
2 mean(vec)
```

Generating sequences in R

- Method 1: Manually write out sequence using `c()`

```
1 ## Basic
2 c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

[1] 1 2 3 4 5 6 7 8 9 10
```

- Method 2: Colon operator (`:`), creates sequences with increments of 1

```
1 c(1:10)

[1] 1 2 3 4 5 6 7 8 9 10
```

- Method 3: `seq()` Function: More flexible and allows you to specify the **start**, **end**, and **by** parameters.

```
1 ## seq 1-10, by = 2
2 seq(1, 10, by = 2)

[1] 1 3 5 7 9
```

Functions

- Function: Input arguments, performs operations on them, and returns a result
- For each of the below functions, what are the:
 - Input arguments?
 - Operations performed?
 - Results?

```
1 ## hint: rnorm simulates random draws from a standard normal distribution
2 random_draws <- rnorm(n = 5,
3     mean = 0,
4     sd = 1)
5
6 ## find the mean
7 mean(random_draws)
```

```
[1] 0.02097222
```

```
1 ## find the median
2 median(random_draws)
```

```
[1] -0.103483
```

```
1 ## find the standard deviation
2 sd(random_draws)
```

```
[1] 0.7640182
```

Keyboard shortcuts

Insert new code cell

- macOS: `Cmd` + `Option` + `I`
- Windows/Linux: `Ctrl` + `Alt` + `I`

Run full code cell or script

- macOS: `Cmd` + `Shift` + `Enter`
- Windows/Linux: `Ctrl` + `Shift` + `enter`

Assignment operator (creates `<-`)

- macOS: `option` + `-`
- Windows/Linux: `option` + `-`

Live coding demo

- Assignment (e.g., `x <- 4`)
- Logical expressions (e.g., `x > 10`)
- Creating a basic sequence
- Your turn next...

In-class exercise 1

1. Assign `x` and `y` to take values 3 and 4.
2. Assign `z` as the product of `x` and `y`.
3. Write code to calculate the square of 3. Assign this to a variable `three_squared`.
4. Write a logical expression to check if `three_squared` is greater than 10.
5. Write a logical expression testing whether `x` is *not* greater than 10. Use the `negate` symbol (`!`).

Exercise 1 solutions

1. Assign `x` and `y` to take values 3 and 4.

```
1 x <- 3
2 y <- 4
```

2. Assign `z` as the product of `x` and `y`.

```
1 z <- x * y
```

3. Calculate the square of 3 and assign it to a variable called `three_squared`.

```
1 three_squared <- 3^2
```

4. Write a logical expression to check if `three_squared` is greater than 10.

```
1 three_squared > 10
```

```
[1] FALSE
```

5. Write a logical expression to test whether `three_squared` is *not* greater than 10. Use the `negate` symbol (`!`).

```
1 !three_squared > 10
```

```
[1] TRUE
```

In-class exercise 2

1. Generate vectors containing the numbers 100, 101, 102, 103, 104, and 105 using 3 different methods (e.g., `c()`, `seq()`, `:`). In what scenarios might each method be most convenient?
2. Generate a sequences of all **even** numbers between 0 and 100. Use the `seq()` function.
3. Create a descending sequence of numbers from 100 to 1, and assign it to a variable. Use the `seq()` function.

Exercise 2 solutions

1. Generate vectors containing the numbers 100 to 105 using three different methods (`c()`, `seq()`, `:`). Discuss the convenience of each method.

```
1 # Generate a vector using c() method
2 vector_c <- c(100, 101, 102, 103, 104, 105)
3
4 # Generate a vector using seq() method
5 vector_seq <- seq(100, 105, by = 1)
6
7 # Generate a vector using : operator
8 vector_colon <- c(100:105)
```

2. Generate a sequence of all even numbers between 0 and 100. Use the `seq()` function.

```
1 # Generate a sequence of all even numbers between 0 and 100
2 even_seq <- seq(0, 100, by = 2)
```

3. Create a descending sequence of numbers from 100 to 1, and assign it to a variable. Use the `seq()` function.

```
1 # Create a descending sequence of numbers from 100 to 1
2 desc_seq <- seq(100, 1, by = -1)
```

Module 3

Working with **vectors** and **data frames**

Learning objectives

- Select elements from **vectors** and columns from **data frames**
- Subset **data frames**
- Investigate characteristics of **data frames**

Indexing vectors

- Basic indexing

```
1 vec <- c(1, 2, 3, 4, 5)
2 first_element <- vec[1]
3 first_element
```

```
[1] 1
```

```
1 third_element <- vec[3]
2 third_element
```

```
[1] 3
```

- Conditional indexing

```
1 vec <- seq(5, 33, by = 2)
2 vec[vec > 25]
```

```
[1] 27 29 31 33
```

Working with data frames

- Data frames are the most common and versatile data structure in R
- Structured as rows (observations) and columns (variables)

```
1 test_scores <- data.frame(  
2   id = c(1, 2, 3, 4, 5),  
3   name = c("Alice", "Bob", "Carol", "Dave", "Emily"),  
4   age = c(25, 30, 22, 28, 24),  
5   gender = c("F", "M", "F", "M", "F"),  
6   score = c(90, 85, 88, 92, 89)  
7 )  
8  
9 knitr::kable(test_scores)
```

id	name	age	gender	score
1	Alice	25	F	90
2	Bob	30	M	85
3	Carol	22	F	88
4	Dave	28	M	92
5	Emily	24	F	89

Working with data frames

- `head()` - looks at top rows of the data frame
- `$` operator - access a column as a vector

```
1 ## print first two rows first row
2 head(test_scores, 2)
```

	id	name	age	gender	score
1	1	Alice	25	F	90
2	2	Bob	30	M	85

```
1 ## access name column
2 test_scores$name
```

```
[1] "Alice" "Bob"   "Carol" "Dave"  "Emily"
```

Subsetting data frames

- Methods:
 - `$`: Single column by name.
 - `df[i, j]`: Row `i` and column `j`.
 - `df[i:j, k:l]`: Rows `i` to `j` and columns `k` to `l`.
- Conditional Subsetting: `df[df$age > 25, ]`.

```
1 ## all rows, columns 1-3
2 test_scores[,1:3]
```

	id	name	age
1	1	Alice	25
2	2	Bob	30
3	3	Carol	22
4	4	Dave	28
5	5	Emily	24

```
1 ## all columns, rows 4-5
2 test_scores[4:5,]
```

	id	name	age	gender	score
4	4	Dave	28	M	92
5	5	Emily	24	F	89

Quiz

Which rows and will this return?

```
1 test_scores[1:3, ]
```

- Which rows and which columns will this return?

```
1 test_scores[test_scores$score >= 90, ]
```

Answers

```
1 test_scores[1:3,]
```

	id	name	age	gender	score
1	1	Alice	25	F	90
2	2	Bob	30	M	85
3	3	Carol	22	F	88

```
1 test_scores[test_scores$score >= 90, ]
```

	id	name	age	gender	score
1	1	Alice	25	F	90
4	4	Dave	28	M	92

Explore **data frame** characteristics

Check number of rows

```
1 ## check number of rows (observations)
2 nrow(test_scores)
```

```
[1] 5
```

Check number of columns

```
1 ## check number of columns (variables)
2 ncol(test_scores)
```

```
[1] 5
```

Check column names

```
1 names(test_scores)
```

```
[1] "id"      "name"    "age"     "gender"  "score"
```

Live coding demo

- Generate random draws from a normal distribution using the `rnorm` function
- Subset the vector of random draws to only include certain observations
- Look at basic summary statistics

In-class exercise 3

1. Generate a **vector** of 100 observations drawn from a normal distribution with a mean of 10 and a standard deviation of 2. Use the **rnorm** function.
2. What are the 1st, 10th, and 100th elements of this **vector**?
3. Calculate the mean of this **vector**. How does this **sample** mean relate to the **population** mean (hint: population mean = 10) of the distribution?
4. Calculate the difference between the sample mean and the population mean. Discuss the reason for the discrepancy.
5. Repeat steps 1-4 with a new sample size of 10,000. Did the difference between the sample mean and the population mean decrease? Why?

Exercise 3 solutions

```
1 # Generate a sample of 100 draws from a normal distribution with mean = 10 and sd = 2
2 sample_data_100 <- rnorm(100,
3                           mean = 10,
4                           sd = 2)
```

```
1 ## look at 1st, 10th, and 100th element
2 sample_data_100[c(1, 10, 100)]
```

```
[1]  8.783910 10.922662  6.510524
```

```
1 # Calculate the mean of this sample
2 sample_mean_100 <- mean(sample_data_100)
3
4 # Calculate the difference between the sample mean and the expected value of the mean
5 difference_100 <- abs(sample_mean_100 - 10)
6
7 difference_100
```

```
[1] 0.0169472
```

Exercise 3 solutions (continued)

```
1 ## Repeat with 10,000 draws
2 sample_data_10000 <- rnorm(10000,
3                             mean = 10,
4                             sd = 2)
5
6 ## look at 1st, 10th, and 100th element
7 sample_data_10000[c(1, 10, 100)]
```

```
[1] 10.048471  7.411483  7.333268
```

```
1 ## mean of this sample
2 sample_mean_10000 <- mean(sample_data_10000)
3
4 ## difference between the sample mean and the population mean
5 difference_10000 <- abs(sample_mean_10000 - 10)
6
7 difference_10000
```

```
[1] 0.002447346
```

Questions?

- Thanks for your attendance and participation
- Please independently complete all exercises in problem set 1 (and review solutions)
- Questions: aasli.nur@sociology.ox.ac.uk